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Targeting ribonucleotide reductase to enhance the antileukemic activity of gilteritinib against chemoresistant *FLT3*- internal tandem duplication acute myeloid leukemia

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Conflict of interest

Yubin Ge is an associate editor of ASPET Discovery. All other authors declare no conflicts of interest. Given his role as associate editor, he had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor.

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Abstract

Acute myeloid leukemia (AML) cases harboring *FMS-like tyrosine kinase 3* (*FLT3*) internal tandem duplication (*FLT3*-ITD) mutations have poor clinical outcomes. Gilteritinib is a United States Food and Drug Administration–approved FLT3 inhibitor for treating relapsed/refractory (R/R) *FLT3*-mutated AML; however, monotherapy shows short-lived responses, highlighting the need for combination therapies. Increased ribonucleotide reductase regulatory subunit M2 (RRM2) was detected in cytarabine-resistant (AraC-R) *FLT3*-ITD AML cell lines and patient-derived xenograft (PDX) cells, accompanied by increased dNDPs determined by proteomics, western blotting, and metabolomics studies. shRNA knockdown of *RRM2* significantly enhanced cell death induced by the ribonucleotide reductase inhibitor, hydroxyurea (HU). Treatment of MV4–11/AraC-R and the PDX cells with variable concentrations of gilteritinib (25–1000 nM) almost completely abolished RRM2 in the cells even at the lowest concentration and was accompanied by a plateau of cell death. Pretreatment with HU (12.5–500 μM) for 48 hours followed by gilteritinib (25–100 nM) for another 24 hours had a strong synergistic effect on the AraC-R *FLT3*-ITD AML cell lines and the PDX cells. Increasing gilteritinib concentrations by 10-fold did not result in further increased cell death. HU treatment induced RRM2 which was abolished by gilteritinib treatment, whereas gilteritinib treatment induced FLT3 which was canceled by HU, demonstrating reciprocal overcoming of drug resistance. Given that both drugs are Food and Drug Administration approved, and HU is well tolerated in both pediatric and adult populations as well as being cost-effective, establishing in vivo models could pave the path to clinical trials, ultimately providing a bridge to transplantation for R/R *FLT3*-ITD AML while minimizing toxicity.

Keywords

Acute myeloid leukemia; Hydroxyurea; Gilteritinib; Cytarabine-resistant; *FMS-like tyrosine kinase 3*-internal tandem duplication

1. Introduction

Acute myeloid leukemia (AML) is a hematological malignancy that is rapidly progressive and aggressive. *FMS-like tyrosine kinase 3* (*FLT3*) internal tandem duplication (*FLT3*-ITD) mutations are found in approximately 25% of AML patients, whereas *FLT3* tyrosine kinase domain (*FLT3*-TKD) mutations are found in about 6%–8% of AML patients.[1] FLT3 is a receptor tyrosine kinase (RTK) that is expressed on CD34+ hematopoietic stem cells and early progenitors and regulates their normal growth and development.[2] Mutations in *FLT3* lead to constitutive activation of the FLT3 RTK, triggering a cascade of signaling events that drive uncontrolled cell proliferation.[3] These mutations are characterized by an allelic ratio of >0.4 mutant *FLT3*-ITD to wild-type *FLT3*. Historically, such mutations have resulted in an overall survival (OS) rate of approximately 20%–30% when treated solely with conventional chemotherapy.[4] Hence, pharmacologic inhibition of the FLT3 RTK via FLT3 inhibitors has emerged as a viable treatment strategy in adults and children.

Gilteritinib is a highly selective oral FLT3 inhibitor that is approved by the US Food and Drug Administration (FDA) for treating relapsed/refractory (R/R) *FLT3*-mutated adult AML patients.[5] Gilteritinib treatment has improved the median OS of R/R *FLT3*-mutated AML patients from 5.6 months to 9.3.[6] Though, even with this advancement the effects are short-lived, highlighting the need for a better understanding of resistance to gilteritinib which is essential for the design of combination therapies.[7,8]

In this study, we show that ribonucleotide reductase (RNR), the rate-limiting enzyme in the biosynthesis of deoxynucleotides, is overexpressed in *FLT3*-ITD AML cell lines with acquired resistance to cytarabine (AraC) compared to the parental cells. We found that combination of hydroxyurea (HU), an oral United States FDA–approved RNR inhibitor that is currently used to treat AML patients with hyperleukocytosis, synergizes with gilteritinib in inducing AraC-resistant (AraC-R) AML cell death. HU treatment induces RNR regulatory subunit M2 (RRM2) which is abolished by gilteritinib treatment. In a similar manner, gilteritinib treatment induces FLT3 which is lowered by HU. These findings demonstrate reciprocal overcoming of drug resistance and support further development of the combination of HU and gilteritinib for the treatment of R/R *FLT3*-ITD AML.

2. Materials and methods

2.1. Drugs

AraC and Z-VAD-FMK were purchased from AbMole. Gilteritinib (Gilt) was purchased from Selleck Chemicals. HU was purchased from MilliporeSigma. 10058-F4 was purchased from MedChemExpress.

2.2. Cell lines

MV4–11, U-937, and HL-60 cell lines were purchased from the American Type Culture Collection (ATCC). MOLM-13 cells were purchased from AddexBio. Cell lines were cultured using RPMI 1640 media plus 10%–20% fetal bovine serum (Thermo Fisher Scientific), 2-mM L-glutamine, and 100 U/mL penicillin and 100 μg/mL streptomycin and incubated in a humidified, 5% CO_2/95 % air environment at 37 °C. Authentication of cell lines was done through the Genomics Core at Karmanos Cancer Institute using the PowerPlex 16 System from Promega. All lines were tested for mycoplasma contamination monthly. MV4–11, MOLM-13, U-937 and HL-60 cells were treated with stepwise increasing concentrations of AraC to generate cells with acquired AraC resistance (designated MV4–11/AraC-R, MOLM-13/AraC-R, U-937/AraC-R and HL-60/AraC-R, respectively), as previously described.[9–11] These resistant cell lines were maintained in AraC-containing media (1100 nM, 1000 nM, 350 nM, and 600 nM, respectively). Cells were drug free for at least 72 hours prior to experimentation.

2.3. Patient-derived xenograft model

A Patient-Derived Xenograft (PDX) model, J000106565, was purchased from Jackson Laboratory. This relapsed model is positive for *FLT3*-ITD, *FLT3*-TKD, and *NPM1* mutations, classified as AML M4/M5, and derived from a patient at relapse who had undergone induction chemotherapy and consolidation therapy with high-dose AraC and allogenic hematopoietic stem cell transplant. J000106565 cells were passaged in NSG-SGM3 female mice (NSGS, JAX#013062; nonobese diabetic severe combined immunodeficient gamma (NOD.Cg-*Prkdc*^{scid} *Il2rg*^{tm1Wjl} 251 Tg(CMV-IL3, CSF2, KITLG)1Eav/MloySzJ (Jackson Laboratory) female mice, as previously described.[12] All mice were provided food and water ad libitum, given supportive fluids and supplements as needed, and housed within an AAALAC accredited animal facility with 24/7 veterinary care. In vivo cell passage was approved by the Institutional Animal Care and Use Committee at Wayne State University.

2.4. Quantification of gene expression by real-time RT-PCR

Total RNA was extracted using TRIzol (Thermo Fisher Scientific) and cDNAs were prepared from 2 μg total RNA using random hexamer primers and a reverse transcription polymerase chain reaction (RT-PCR) kit (Life Technologies), and purified using the QIAquick PCR Purification Kit (Qiagen), as previously described.[13–15] *RRM2* (Hs00357247_g1) and *c-Myc* (Hs99999003_m1) and *FLT3* (Hs00174690_m1) transcripts were quantitated using TaqMan probes (Thermo Fisher Scientific) and a LightCycler 480 real-time PCR machine (Roche Diagnostics), based on the manufacturer's instructions. Real-time PCR results were expressed as means from 3 independent experiments and were normalized to *RPL13a* (Hs03043885_g1) transcripts. Fold changes were calculated using the comparative Ct method.[16]

2.5. Proteomics

MV4–11/AraC-R cell line was cultured in media without AraC for 3 days. MV4–11 and MV4–11/AraC-R cells were seeded in fresh media, and 16 hours later the cells were harvested, washed with PBS, and stored as cell pellets at −80 °C until analysis. Global proteomic analysis was performed by the Wayne State University Proteomic Core, as previously reported.[17] Briefly, samples were solubilized in 2% Lithium dodecyl sulfate and heated to 95 °C for 5 minutes then viscosity eliminated by filtering through a spin cartridge. Samples were reduced/alkylated with DTT/IAA prior to methanol precipitation. The precipitated proteins were washed with methanol then resolubilized and digested with trypsin. Peptide abundance was determined using a fluorescent peptide quantitation kit (Thermo Fisher Scientific). Each digest was labeled with a unique TMT-Pro multiplexing reagent then analyzed independently by liquid chromatography-tandem mass spectrometry on an Orbitrap Fusion MS system to establish completeness of labeling. Once >99 % labeling was confirmed, the samples were pooled, and the mixtures fractionated by alkaline reverse-phase spin column (Thermo Scientific Pierce). The fractions were analyzed on the Orbitrap Eclipse using 2-hour cycle time analytical runs. All data were analyzed using Proteome Discoverer 2.5 software. A total of 6 biological replicates per group were analyzed. The levels of protein expression were normalized by median for each sample followed by log2-transformation. The group comparison was performed using a 2-sample *t* test followed by false discovery rate correction. The differentially expressed proteins were identified using 2 cutoffs, a false discovery rate of ≤5 % and fold-change of ≥1.2.

2.6. Targeted metabolomics

MV4–11/AraC-R cell line was cultured in media without AraC for 3 days. MV4–11 and MV4–11/AraC-R cells were seeded in fresh media, and 16 hours later the cells were harvested, washed with PBS, and then stored at −80 °C until analysis. Metabolomics study was performed by the Pharmacology and Metabolomics Core at the Karmanos Cancer Institute and analyzed as previously described.[9,18,19]

2.7. Western blot

Cells were lysed in buffer with protease and phosphatase inhibitors (Roche Diagnostics) and sonicated as described.[20] Whole-cell lysates underwent SDS-PAGE were transferred onto polyvinylidene difluoride membranes (Thermo Fisher Scientific) and probed with anti-RRM1 (ab137114, Abcam), anti-RRM2 (ab172476, Abcam), anti-cleaved caspase 3 (9661S, Cell Signaling Technology), anti-c-Myc (5605S, Cell Signaling Technology), anti-FLT3 (ab245116, Abcam), or anti-ß-actin (MilliporeSigma) antibody. Proteins were visualized using the Odyssey Infrared Imaging System (LI-COR). All blots were repeated in triplicate and densitometry was performed with Image Studio Software (v4.0.21, LI-COR) and normalized to ß-actin.

2.8. DepMap: public data set analysis

Publicly available dependency data were obtained from the Cancer Dependency Maps (DepMap) portal (https://depmap.org/portal/, version 12.16.2). DepMap uses CRISPR knockout screening to determine dependency scores (chronos) of cancer cell lines on genes.[21] The Public 24Q2 dataset was employed to evaluate dependency of AML cell lines on *RRM2* gene.

2.9. Annexin V/PI staining and flow cytometry analysis

Cells post drug treatments for 24–72 hours were collected, washed with PBS, and resuspended in 1x Annexin V Binding Buffer (556454, BD Biosciences), and then stained in triplicate with 5 μL propidium iodine (PI, 556463, BD Biosciences) and 5 μL annexin V-fluorescein isothiocyanate (annexin V-FITC, 640945, BioLegend). Flow cytometry was performed using the Cytek Northern Lights at the Microscopy, Imaging, and Cytometry Resources Core at the Karmanos Cancer Institute. A fixed cell volume was analyzed for annexin V+ and viable cells (annexin V−/PI−). Data were quantified using FlowJo (v10.10.0, BD Biosciences) to distinguish viable (unstained), early apoptotic (annexin V+/PI−), and late apoptotic/dead (annexin V+/PI+) cells. Combination index (CI) values were calculated with CompuSyn (Combosyn Inc) to determine synergy (CI < 1), additivity (CI = 1), or antagonism (CI > 1).

2.10. Lentiviral knockdown of RRM2

The pMD-VSV-G and delta 8.2 plasmids were gifts from Dr Dong at Tulane University. *RRM2* (TRCN0000038962) and nontarget control (NTC, SHC002V) shRNA lentiviral constructs were purchased from MilliporeSigma. Lentivirus production and transduction were carried out as previously described.[22,23]

2.11. Statistical analysis

Unpaired *t* tests were used for comparisons between 2 different treatment groups. One-way analysis of variance followed by Tukey's post hoc test was used when comparing more than 2 groups. GraphPad Prism 9.0 was used to perform statistical analyses. Error bars represent the standard error of the mean. Statistical significance was set at $P < .05$.

3. Results

3.1. RRM2 protein levels are significantly increased in AraC-resistant FLT3-ITD AML cells

Given the role of nucleotide metabolism in chemotherapy response, we investigated whether dysregulation of RNR contributes to AraC resistance in *FLT3*-ITD AML. Proteomic profiling of the parental *FLT3*-ITD AML cell line MV4–11 and its AraC-resistant derivative (MV4–11/AraC-R) revealed significant upregulation of RRM2 (Fig. 1A). Western blot validation confirmed this pattern, showing significantly elevated RRM2, along with RRM1, protein levels in the resistant cells cultured in AraC-containing media or AraC-free media for 72 hours, compared to the parental MV4–11 cells (Fig. 1B). Functionally, metabolomic profiling revealed significant increase of intracellular dNDP levels in the resistant cells (Fig. 1C), compared to the parental cells, supporting the hypothesis that elevated RRM2 drives enhanced nucleotide biosynthesis in the resistant cells. Further supporting this hypothesis, we observed similar patterns of RRM2 and RRM1 increase in AraC-resistant MOLM-13 cells (MOLM-13/AraC-R), compared to the parental MOLM-13 cells, further reinforcing this mechanism (Fig. 1D).

To validate these findings, we used the J000106565 PDX model, established from a *FLT3*-ITD AML patient relapsed post chemotherapy and post bone marrow transplant. Spleen cells from NSGS mice injected with J000106565 (>95% human AML cells) were analyzed for RNR subunit levels. Consistent with results in the AraC-R cell lines, RRM2 levels were elevated in the J000106565 PDX cells compared to the MOLM-13/AraC-R cells, whereas RRM1 levels were similar compared to the MOLM-13/AraC-R cells (Fig. 1E). Interestingly, AraC-R AML cell lines lacking the *FLT3*-ITD mutation (HL-60/AraC-R and U-937/AraC-R) did not show consistent RRM2 upregulation compared to the parental lines, suggesting that RRM2 upregulation may be linked to FLT3-ITD–driven signaling pathways (Fig. 1F), though upregulation in the FLT3 wild-type HL60/AraC-R cells suggests that a potential alternative mechanism exists. Taken together, these findings support the conclusion that RRM2 is a primary driver of enhanced nucleotide synthesis in AraC-resistant *FLT3*-ITD AML and may represent a potential vulnerability for treating relapsed *FLT3*-ITD AML.

3.2. Downregulation of RRM2 sensitizes AraC-resistant FLT3-ITD AML cells to hydroxyurea

Given the elevated levels of RRM2 in AraC-resistant *FLT3*-ITD AML cells, we next sought to determine whether RRM2 represents a functional dependency and potential therapeutic target in these cells. Data from the DepMap portal indicates that *RRM2* is an essential gene for AML cell survival, further supporting its therapeutic relevance (Fig. 2A). Pharmacological inhibition of RRM2 with HU led to the induction of cell death and decrease of viable cells in MV4–11/AraC-R, MOLM-13/AraC-R, and J000106565 PDX cells. MOLM13/Ara-C-R cells had the strongest response to HU followed by J000106565 PDX cells and MV4–11/AraC-R cells. Correspondingly, HU treatment led to a significant reduction in viable cells (Fig. 2B-C). It is important to note that the levels of apoptosis induced by HU seem to negatively associate with the levels of RRM2 in these AraC-resistant cells, indicating high levels of RRM2 cause resistance to HU. Together, these results demonstrate that AraC-resistant *FLT3*-ITD AML cells respond to RNR inhibition, supporting RRM2 as a potential therapeutic target in the resistant disease.

To directly assess whether RRM2 levels influence HU sensitivity, we performed lentiviral knockdown of *RRM2* in MV4–11/AraC-R cells. Infection with RRM2-targeting shRNA resulted in ~60% reduction of RRM2 protein levels in the cells (designated *RRM2* KD cells) compared to cells infected with the non-targeting control (designated NTC cells, Fig. 2D, E). The *RRM2* KD and NTC cells were then treated with increasing concentrations of HU. Annexin V/PI staining and flow cytometry analyses revealed enhanced cell death and greater suppression of viable cells in the *RRM2* KD cells compared to the NTC controls (Fig. 2F, G). These findings suggest that RRM2 levels are a determinant of sensitivity to HU for AraC-resistant *FLT3*-ITD AML cells, and that suppression of RRM2 can partially sensitize resistant cells to HU.

3.3. Antileukemic activity of gilteritinib and its connection to RRM2

Building on these findings, we next explored whether FLT3 inhibitor, gilteritinib, modulated RRM2 protein levels as part of its antileukemic activity in targeting FLT3-ITD signaling. One potential link is c-Myc, a transcription factor in the Myc family, that has been shown to be a major target of FLT3-ITD in AML cells[24] and our previous study showed that gilteritinib substantially downregulates c-Myc in *FLT3*-ITD AML cells.[9] In addition, targeting c-Myc results in decrease of RRM2.[25] Thus, RRM2 may lie downstream of FLT3-ITD/c-Myc signaling, providing a mechanistic rationale for targeting this axis to suppress RRM2-driven resistance to HU. Annexin V/PI staining and flow cytometry analysis following treatment with increasing concentrations of gilteritinib (25–1000 nM) revealed that cell death induction plateaued around 25–50 nM gilteritinib in the AraC-R *FLT3*-ITD AML cell lines and J000106565 PDX cells (Fig. 3A). Treatment also resulted in similar decrease of viable cells across all concentrations (Fig. 3B). Mechanistically, 25 nM of gilteritinib almost completely abolished c-Myc and RRM2 in MV411/AraC-R and J000106565 PDX cells. RRM1 remained unchanged by gilteritinib treatment confirming RRM2 as the key subunit in promoting cell survival (Fig. 3C). Treatment of MV4–11/AraC-R and J000106565 PDX cells with 10058-F4, a c-Myc inhibitor, resulted in caspase 3 cleavage/activation and decrease of c-Myc and RRM2 protein levels, which could not be rescued by Z-VAD-FMK (Z-VAD), a pan-caspase inhibitor (Fig. 3D). These findings confirm that gilteritinib downregulates RRM2 mediated by c-Myc in AraC-resistant *FLT3*-ITD AML cells, supporting RRM2 as a key effector of FLT3-ITD–driven survival of these leukemic cells.

3.4. Gilteritinib and hydroxyurea synergistically induce cell death in MV4–11/AraC-R, MOLM-13/AraC-R and J000106565 PDX cells

Collectively, our data support a model in which FLT3-ITD–driven c-Myc signaling promotes leukemic cell survival through regulation of RRM2, and that targeting RRM2 may represent an effective strategy to overcome resistance to gilteritinib in *FLT3*-ITD AML. However, the cooperation if not synergy between HU and gilteritinib could be dependent on the sequence of drug administration. To test this hypothesis, we tested the antileukemic interactions between HU and gilteritinib at 3 different treatment schedules. Cells were treated for a determined duration of time, followed by annexin V/PI staining and flow cytometry analysis. CompuSyn software was used to calculate CI values to determine synergy. MV4–11/AraC-R cells exposed to HU + gilteritinib simultaneously for 72 hours demonstrated a modest synergistic effect when analyzed using annexin V+ cells and viable cells (CI < 0.79 and CI < 0.86, respectively, Fig. 4A–C). A sequential treatment with HU (12.5–500 μM) for 48 hours followed by gilteritinib (25–100 nM) for another 24 hours had a strong synergistic effect (CI < 0.58, Fig. 4D–F). In contrast, sequential treatment with gilteritinib for 24 hours followed by HU for another 48 hours resulted in antagonistic effects (CI > 1) in the cells (Fig. 4 G–I). These results show that sequential treatment with HU for 48 hours followed by gilteritinib for an additional 24 hours is an optimal drug administration sequence to achieve strong synergy.

Strong synergy between HU and gilteritinib was also detected in MOLM-13/AraC-R and J000106565 PDX cells following the optimal drug administration sequence identified above (Fig. 5A–D). Because peak plasma concentrations of gilteritinib at the approved clinical dose can reach up to 1 μM,[26] we tested higher concentrations of gilteritinib in combination with HU. As monotherapy, increasing gilteritinib concentrations in MV4–11/AraC-R and J000106565 PDX cells by 10-fold did not result in further increase of annexin V+ cells (Fig. 5E-H). Taken together, we have identified that sequential treatment with HU for 48 hours followed by gilteritinib results in a strong synergistic effect in AraC-R *FLT3*-ITD AML cells.

3.5. Mechanistic insights into combination treatment with HU and gilteritinib

Using the 48-hour HU followed by additional 24 hours with gilteritinib treatment schedule, we lastly investigated this combination effect on RNR, c-Myc and FLT3 protein levels. Western blot analysis demonstrated an induction of RRM2 by HU which was abolished by gilteritinib treatment; similarly, gilteritinib treatment induced FLT3 which was attenuated by HU (Fig. 6A). *c-Myc* mRNA levels were significantly reduced following treatment with HU, gilteritinib, and HU + gilteritinib. *RRM2* mRNA levels were significantly increased following HU treatment, while gilteritinib treatment either significantly decreased *RRM2* transcript levels in one cell line and had no significant effect on the mRNA levels in the other. *FLT3* mRNA levels were significantly decreased following HU treatment and increased following gilteritinib treatment. *RRM2* and *FLT3* mRNA levels were significantly elevated following combination treatment, though protein levels were reduced (Fig. 6A), suggesting that posttranscriptional mechanisms contribute to RRM2 and FLT3 regulation under the combination treatment.

4. Discussion

Targeted therapies have significantly improved treatment outcomes for patients with R/R AML, particularly those harboring *FLT3*-ITD mutations. Our study provides mechanistic insights and shows synergistic antileukemic interaction for a novel combination strategy using 2 FDA-approved agents, HU and gilteritinib, to overcome resistance and improve therapeutic response in AraC-resistant *FLT3*-ITD AML.

There have been many studies that have identified RRM2 as a therapeutic target due to its dysregulation in gastrointestinal, lung, and head/neck cancers.[27,28] Although HU has long been used for cytoreduction in AML, its role in combination therapy has not been extensively explored in the context of *FLT3*-ITD AML. Our results show, for the first time, that RRM2 is upregulated in AraC-resistant *FLT3*-ITD AML cells, and it is essential for the survival of these resistant cells, indicating that downregulation of RRM2 could sensitize these resistant cells to HU. Although the literature and our previous work suggest that targeting the FLT3-ITD/c-Myc signaling pathway could result in downregulation of RRM2,[9,24,29,30] the molecular mechanism has not been defined. Our study shows that gilteritinib effectively attenuates the induction of RRM2 protein by HU (possibly through compensatory effect), providing one molecular basis for the synergistic antileukemic activities of combined gilteritinib and HU against *FLT3*-ITD AML.

It was interesting to find that HU transcriptionally downregulates *FLT3*, though the detailed molecular mechanisms remain to be determined. This property renders HU the capacity to cancel the induction of FLT3-ITD by gilteritinib, providing a second molecular mechanism responsible for the synergy between the 2 agents.

It is important to note that 10-fold reduction in gilteritinib concentration did not result in loss of in vitro efficacy, indicating the potential to reduce treatment-related toxicity such as anemia, hepatic injury, and differentiation syndrome.[31] However, this combination is schedule-dependent as sequential treatment with HU for 48 hours followed by gilteritinib for an additional 24 hours maximizes synergy. It is possible that priming of the cells with HU results in substantially lower levels of FLT3-ITD, rendering the leukemia cells sensitive to lower concentrations of gilteritinib; HU treatment induces RRM2, a mechanism by which AML cells fight for survival from HU treatment, making them more susceptible to RRM2 downregulation induced by gilteritinib. It is likely that treating cells with gilteritinib first completely abolishes RRM2 thereby mitigating any effect HU would have on this target, leading to antagonistic effect. Moving forward, in vivo validation of our in vitro efficacy of the combination is critical. Regardless of this limitation, our findings suggest that this combination has promising translational potential, because both HU and gilteritinib are United States FDA-approved and accessible.

Gilteritinib has shown early promise in pediatric R/R *FLT3*-mutated AML, a patient population with high relapse rates and limited treatment options.[32] In pediatric AML, especially *FLT3*-ITD cases, classified as high-risk, stem cell transplantation is often pursued in first remission.[33] However, for patients with positive minimal residual disease or challenges in donor availability, additional therapies are needed to achieve or maintain remission before transplant.[34] Our data suggest that this combination could serve as a bridge to transplant, keeping leukemic burden low or achieving complete remission. In adult patients not eligible for transplant, this strategy may also provide an alternative to gilteritinib monotherapy, which yields a median OS of only ~9 months.[6] By enhancing efficacy while minimizing toxicity, HU + gilteritinib combination therapy could offer a more effective and better-tolerated option for high-risk, relapsed *FLT3*-ITD AML.

5. Conclusions

In summary, we determined the combination of HU and gilteritinib has synergistic antileukemic activity against AraC-resistant *FLT3*-ITD AML at clinically achievable concentrations. Our preclinical data, along with the demonstrated safety and efficacy of both HU and gilteritinib in AML, support further investigation of this combination therapy for treatment for pediatric and adult R/R *FLT3*-ITD AML.

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Data availability

The authors declare that all the data supporting the findings of this study are contained within the paper.

Abbreviations

AML	acute myeloid leukemia
AraC	cytarabine
AraC-R	cytarabine-resistant
CI	combination index
FDA	Food and Drug Administration
FLT3	FMS-like tyrosine kinase 3
***FLT3*-ITD**	FLT3-internal tandem duplication
Gilt	gilteritinib
HU	hydroxyurea
MOLM-13/AraC-R	AraC-resistant MOLM-13 cells
MV4–11/AraC-R	AraC-resistant MV4–11 cells

NSGS	NSG-SGM3 mice
NTC	nontarget control
OS	overall survival
PDX	patient-derived xenograft
R/R	relapsed/refractory
RNR	ribonucleotide reductase
RRM2	RNR regulatory subunit M2
Z-VAD	Z-VAD-FMK
RTK	receptor tyrosine kinase

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Significance Statement:

This study demonstrates the reciprocal overcoming of drug resistance and supports further development of the combination of hydroxyurea and gilteritinib for the treatment of relapsed/refractory *FMS-like tyrosine kinase 3* internal tandem duplication acute myeloid leukemia. Moreover, this study highlights the importance of drug repurposing.

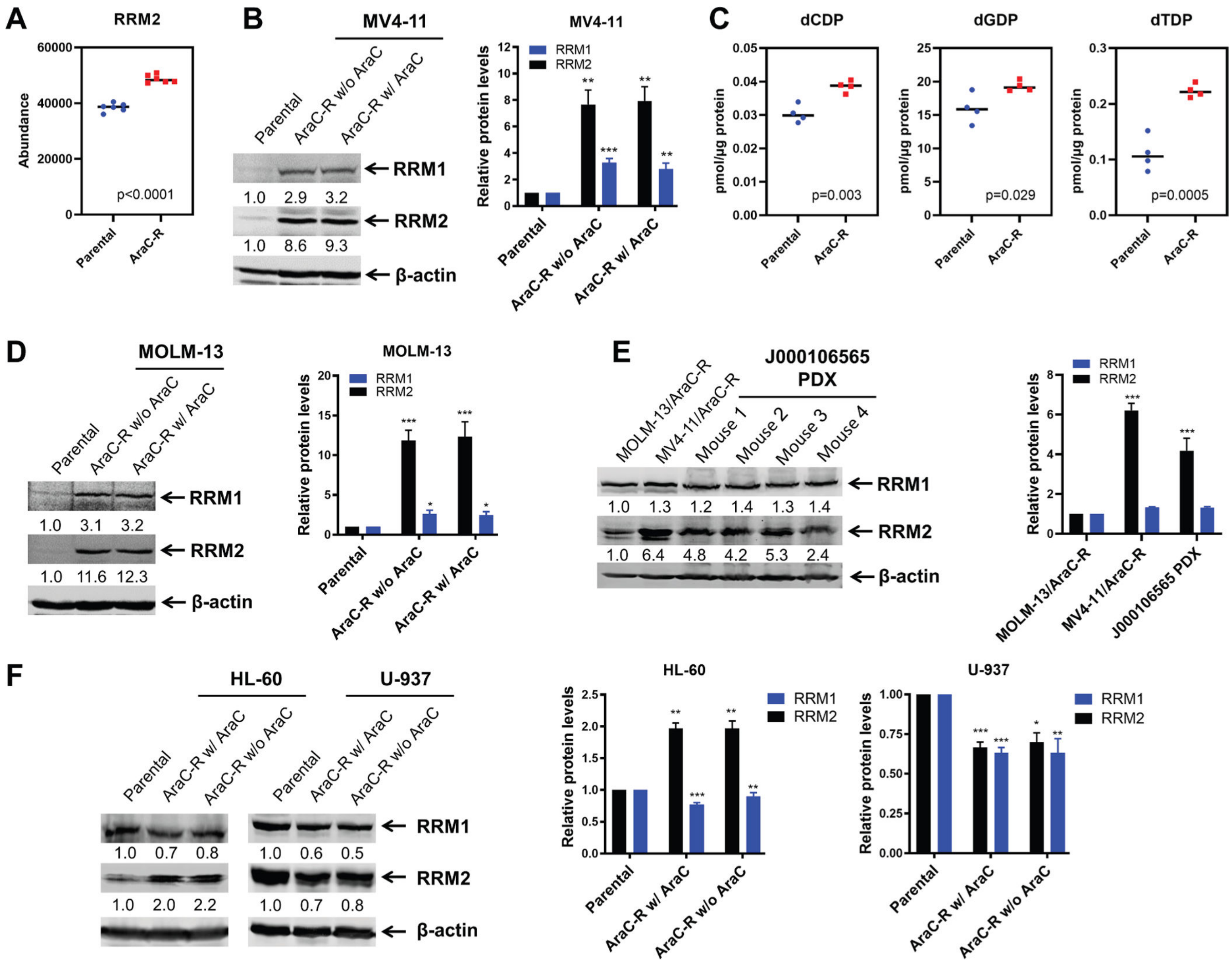

Fig. 1.

Higher levels of RNR in AraC-resistant *FLT3*-ITD AML cells. (A) Levels of intracellular RRM2 between MV4–11 parental and MV4–11 AraC-resistant (MV4–11/AraC-R) cells were compared via proteomics analysis. (B) RRM1 and RRM2 protein levels in parental and AraC-R MV-11 cells, with and without AraC in the culture media for 72 hours before cell lysis. Densitometry measurements, normalized to β-actin then compared to parental cells, are shown below the corresponding blot and replicates are graphed on the right. ** indicates $P < .01$ and *** indicates $P < .001$ compared to the parental cells. (C) Targeted metabolomics data of dCDP, dGDP, and dTDP levels in parental and AraC-R MV4–11 cells. (D, E) RRM1 and RRM2 protein levels in MOLM-13, MOLM-13/AraC-R (with and without AraC in the culture media for 72 hours prior to cell lysis), MV4–11/AraC-R, and in patient-derived xenograft (PDX) cells (J000106555, derived from a *FLT3*-ITD AML patient at relapse post chemotherapy and bone marrow transplant) expanded in NSGS mice. Densitometry measurements, normalized to β-actin then compared to vehicle controls, are shown below the corresponding blot and replicates are graphed on the right. * indicates $P < .05$ and *** indicates $P < .001$ compared to the parental MOLM-13 (panel D) or

MOLM-13/AraC-R cells (panel E). (F) RRM1 and RRM2 protein levels in *FLT3* wild-type parental HL-60 and U-937 and AraC-R (with and without AraC in the culture media 72 hours prior to cell lysis). Densitometry measurements, normalized to β-actin then compared to the parental cells, are shown below the corresponding blot and replicates are graphed on the right. * indicates $P < .05$, ** indicates $P < .01$, and *** indicates $P < .001$ compared to the parental cells.

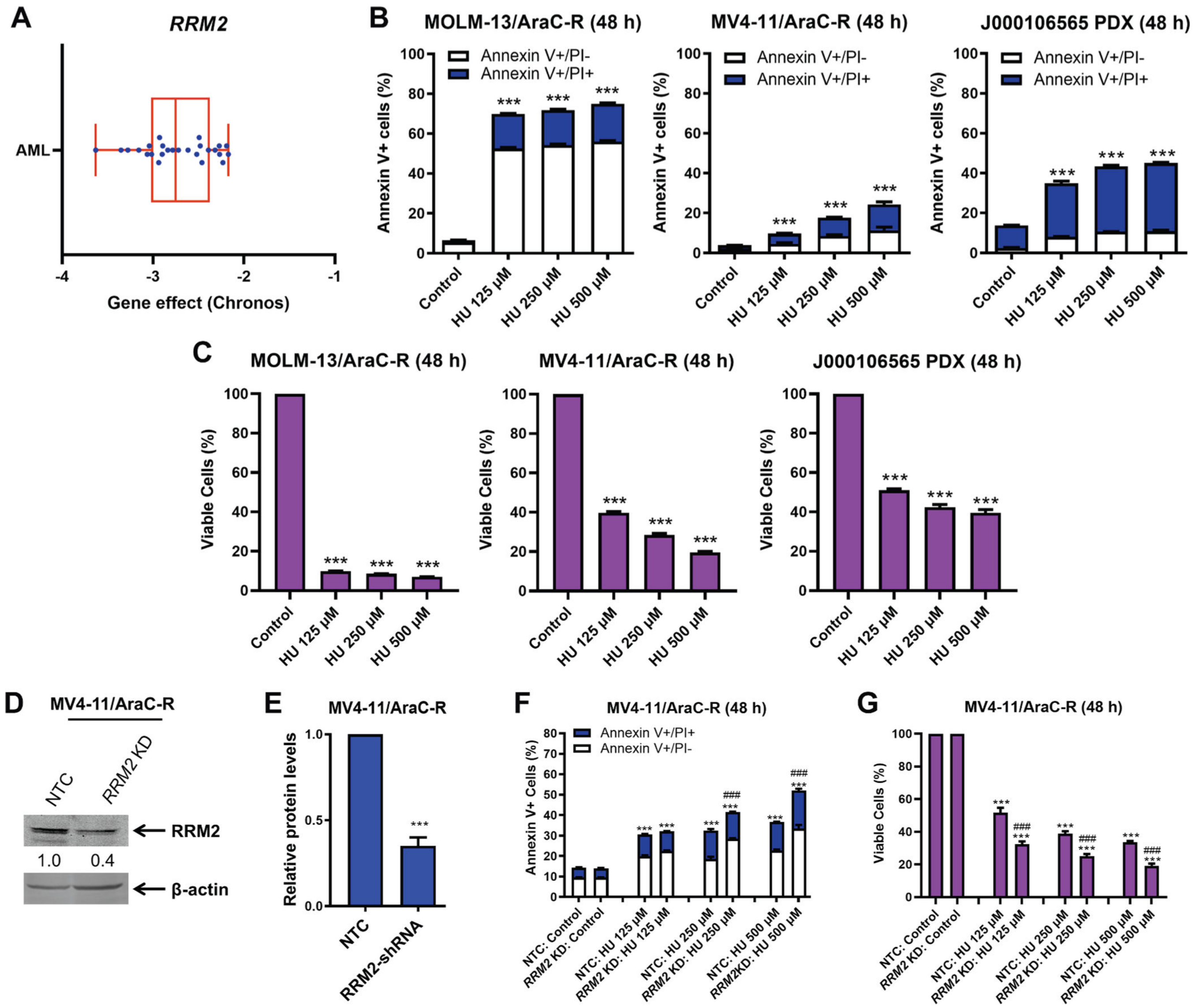

Fig. 2.

Targeting RRM2 with HU induces apoptosis in AraC-R *FLT3*-ITD AML cells. (A) The Chronos scores for *RRM2* across AML cell lines, derived from the DepMap database (Public 24Q2 dataset). (B, C) MOLM-13/AraC-R, MV4–11/AraC-R, and J000106565 PDX cells were treated with variable concentrations of HU for 48 hours. Cells were stained with annexin V-FITC/PI and assessed via flow cytometry analysis. Viable cells (annexin V−/PI−) were calculated and normalized to vehicle control (panel C). *** indicates $P < .001$ compared to vehicle control. (D–G) Lentiviral shRNA knockdown of *RRM2* was performed in MV4–11/AraC-R cells. Nontarget control (NTC) shRNA was used as the control. Whole-cell lysates were subjected to western blotting. The fold changes for the densitometry measurements, normalized to β-actin and compared to NTC, are indicated below the corresponding blot (panel D) and replicates are graphed in panel E. The knockdown cells were treated with HU for 48 hours, followed by annexin V-FITC/PI staining and flow cytometry analysis (panel F). Viable cells (annexin V−/PI−) were calculated and normalized to vehicle control (panel G). *** indicates $P < .001$ compared to control and ### indicates $P < .001$ compared to NTC treated with the same HU concentration.

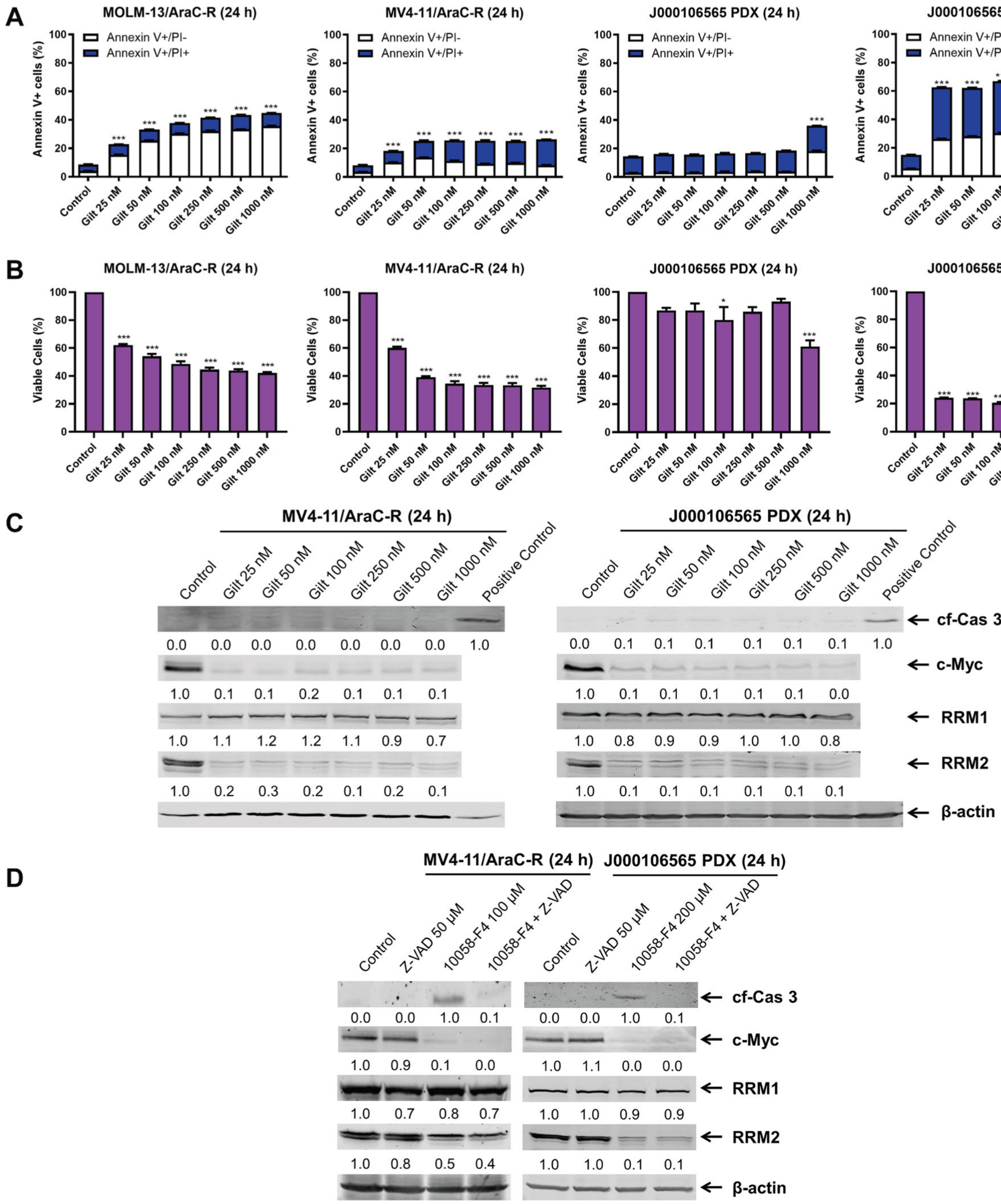

Fig. 3.

Gilteritinib treatment downregulates RRM2 and c-Myc in AraC-R *FLT3*-ITD AML cell lines and PDX cells. (A, B) MOLM-13/AraC-R, MV4–11/AraC-R, and J000106555 PDX cells were treated with varying concentrations of gilteritinib for 24–48 hours and then subjected to annexin V-FITC/PI staining and flow cytometry analysis. Viable cells (annexin V−/PI−) were calculated and normalized to vehicle control (panel B). * indicates $P < .05$ and *** indicates $P < .001$ compared to control. (C) MV4–11/AraC-R and J000106565 PDX cells were treated with varying concentrations of gilteritinib for 24 hours. Cells treated with 500 nM CUDC-907 for 24 hours were used as positive controls. Whole-cell lysates were subjected to western blotting. The fold changes for the densitometry measurements, normalized to β-actin and compared to control, are indicated below the corresponding blot. Cleaved caspase 3 is indicated by cf-Cas 3. (D) MV4–11/AraC-R and J000106565 PDX cells were treated with the pan-caspase inhibitor Z-VAD-FMK (Z-VAD) alone or in combination with the c-Myc inhibitor 10058-F4 for 24 hours. Whole-cell lysates were subjected to western blotting. The fold changes for the densitometry measurements, normalized to β-actin and compared to control, are indicated below the corresponding blot.

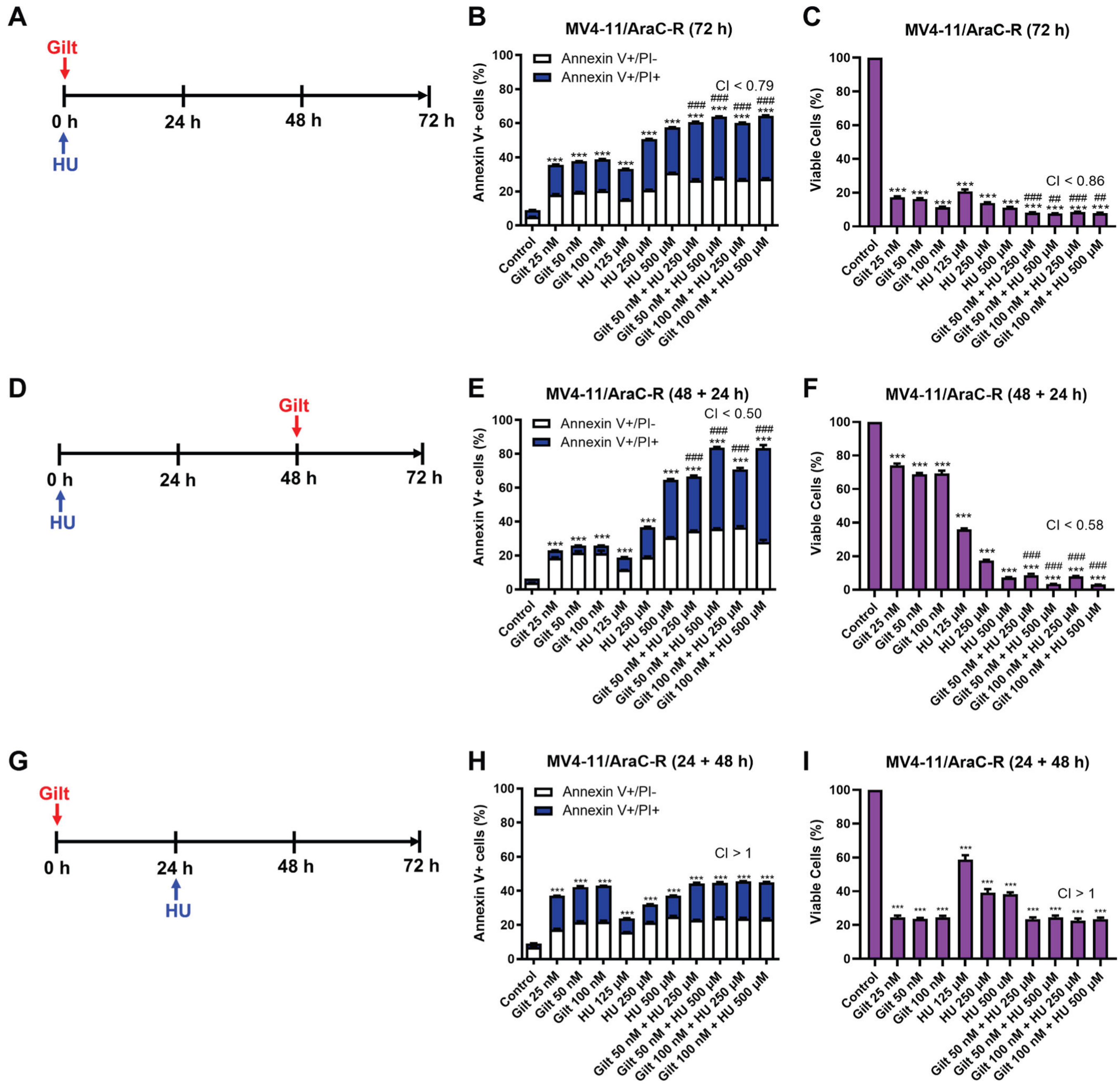

Fig. 4.

Schedule-dependent synergistic induction of apoptosis by gilteritinib and HU in AraC-R *FLT3*-ITD AML cell line MV4–11/AraC-R. MV4–11/AraC-R cells were treated simultaneously with HU and gilteritinib for 72 hours (panels A–C), HU for 48 hours followed by gilteritinib for an additional 24 hours (panels D-F), or gilteritinib for 24 hours followed by HU for an additional 48 hours (panels G–I). Cells were stained with annexin V-FITC/PI and assessed via flow cytometry analysis. Viable cells (annexin V−/PI−) were calculated and normalized to vehicle control (panels C, F, and I). *** indicates $P < .001$ compared to control and ### indicates $P < .001$ compared to single-drug treatments.

Combination index (CI) values were calculated using CompuSyn software to determine synergistic (CI < 1), additive (CI = 1), or antagonistic (CI > 1) effect.

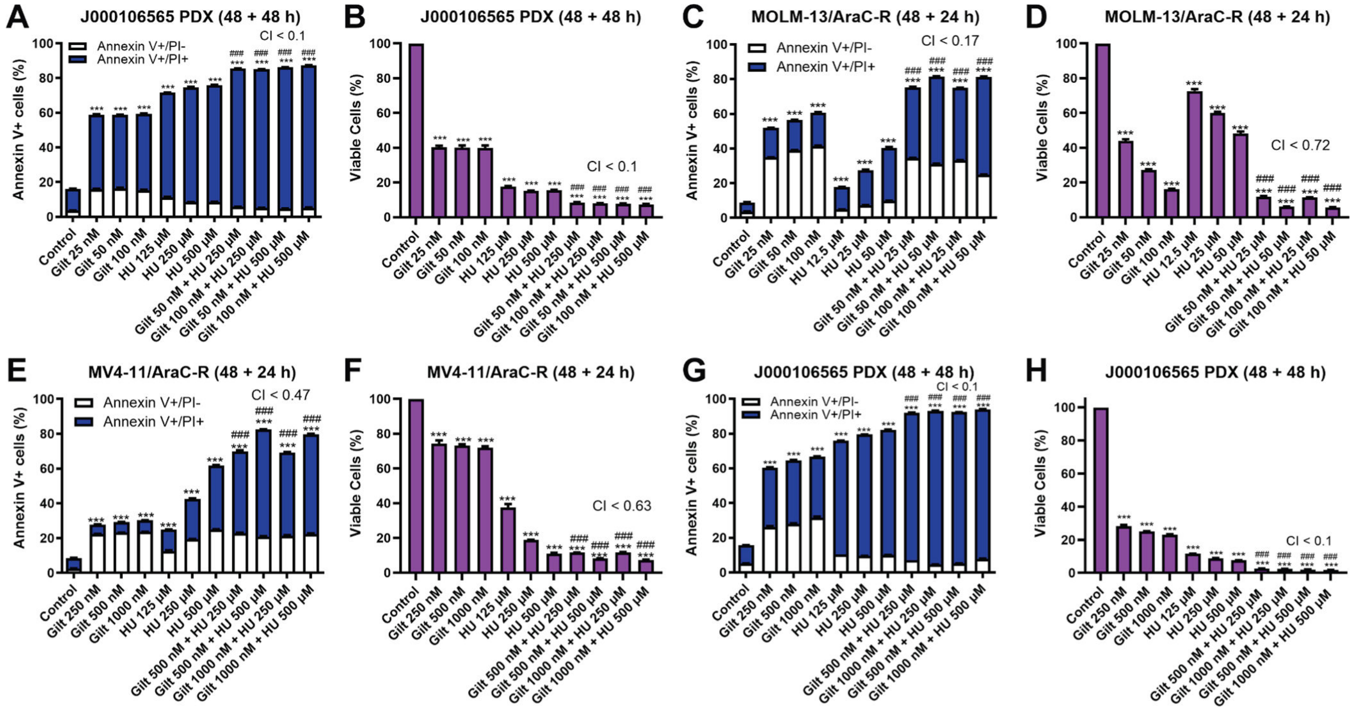

Fig. 5.

Pretreatment with HU synergistically induces apoptosis in AraC-R *FLT3*-ITD AML cell lines and PDX cells. (A-B) J000106565 PDX cells were treated with HU for 48 hours followed by gilteritinib for an additional 48 hours. Cells were stained with annexin V-FITC/PI and assessed via flow cytometry analysis. Viable cells (annexin V−/PI−) were calculated and normalized to vehicle control. CI values were calculated using CompuSyn software. *** indicates $P < .001$ compared to control; ### indicates $P < .001$ compared to single-drug treatments. (C, D) MOLM-13/AraC-R cells were treated with HU for 48 hours followed by gilteritinib for an additional 24 hours. Cells were stained with annexin V-FITC/PI and assessed via flow cytometry analysis. Viable cells (annexin V−/PI−) were calculated and normalized to vehicle control. CI values were calculated using CompuSyn software. *** indicates $P < .001$ compared to control; ### indicates $P < .001$ compared to single-drug treatments. (E–H) MV4–11/AraC-R and J000106565 cells were treated with HU for 48 hours followed by 10-fold higher concentrations of gilteritinib (250–1000 nM) than that shown in Figures 4E and 5A. Cells were stained with annexin V-FITC/PI and assessed via flow cytometry analysis. Viable cells (annexin V−/PI−) were calculated and normalized to vehicle control. CI values were calculated using CompuSyn software to determine synergistic (CI < 1), additive (CI = 1), or antagonistic (CI > 1) effect. *** indicates $P < .001$ compared to control; ### indicates $P < .001$ compared to single-drug treatments.

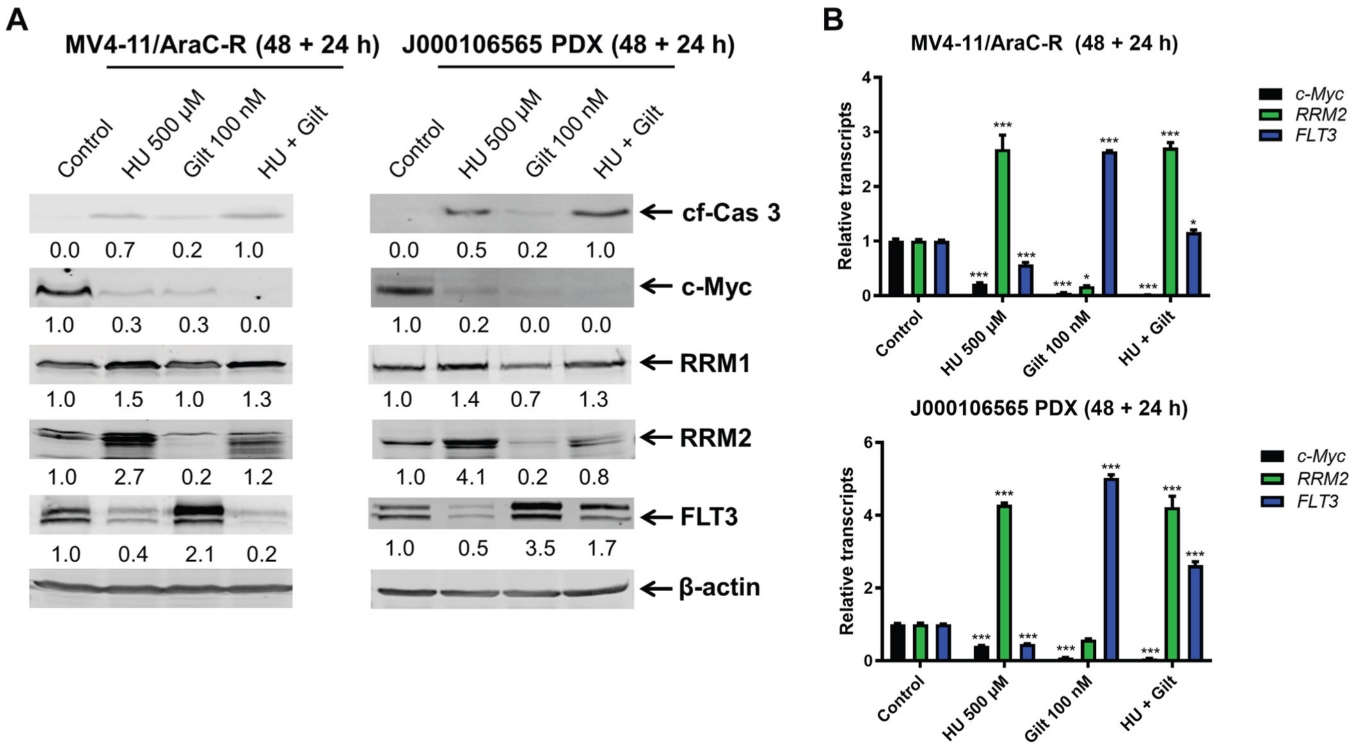

Fig. 6.

Combination treatment with HU + gilteritinib downregulates RRM2, c-Myc, and FLT3 in AraC-R *FLT3*-ITD AML cell lines and PDX cells. (A) MV4–11/AraC-R and J000106565 PDX cells were treated with HU for 48 hours followed by gilteritinib for an additional 24 hours. Whole-cell lysates were subjected to western blotting. The fold changes for the densitometry measurements, normalized to β-actin and compared to control, are indicated below the corresponding blot. (B) MV4–11/AraC-R and J000106565 PDX cells were treated with HU for 48 hours followed by gilteritinib for 24 hours. Total RNA was isolated, and *RRM2*, *c-Myc*, *FLT3*, and *GAPDH* transcripts were measured by reverse transcription polymerase chain reaction. Fold changes were calculated using the comparative Ct method. * indicates $P < .05$ and *** indicates $P < .001$ compared to control.